MIRAGE

Measuring the Impact and Reality of Rainwater Harvesting Mandates

A meta-analysis of rooftop rainwater harvesting performance in Indian cities

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CM617 Sustainability Principles and Practices, IIT Bombay. Pathway 1: Sustainability Research. Sector: Water.

Twenty years of mandates, almost no measurement

2003

Chennai makes RWH compulsory for every building; Bengaluru, Delhi, Mumbai and 20+ others follow

33

states and union territories have adopted the Model Building Bye-Laws 2016 RWH provision

None

consolidated quantitative syntheses of what installed systems deliver that we could find

The gap in the evidence

Mandates are enforced at plan approval: no completion certificate or water connection without a structure. After that, checks are rare and results are not published. Single-building case studies, agency demonstration reports and newspaper audits exist, but we found no study that combines them. MIRAGE collects the published numbers, puts them on one scale and asks what the mandate delivers.

Three questions, four searches

Q1. Performance gap

How much of the theoretical rooftop potential do documented systems actually harvest or recharge?

Q2. Functionality

What share of installed systems stay functional, and why do they fail?

Q3. Cost

What does harvested water cost per kilolitre compared with piped, desalinated and tanker water?

89

performance site records

43

cost records + 37 price benchmarks

46

surveys, audits, inspections; 40 sources state failure reasons

24

mandate instruments

Sources: CGWB compilations and manual, CSE case studies, Lok Sabha replies, Delhi Jal Board and NGT reports, Arghyam, journal papers, campus and company reports, IMD normals, RBI CPI. Every number logged with its URL and the sentence or values taken from the page.

One metric, three evidence tiers

Potential P = Area (m2) x Rainfall (mm) x C / 1000 [kL/yr] Performance ratio = reported volume / P. C = 0.80 (CGWB manual, Table 7.2: concrete 0.7 to GI sheet 0.9); sensitivity 0.70 and 0.90.

Observed

Source names a flow meter, a measuring device at the recharge well or a tank-filling record. 7 observations, 5 sites.

Agency-reported

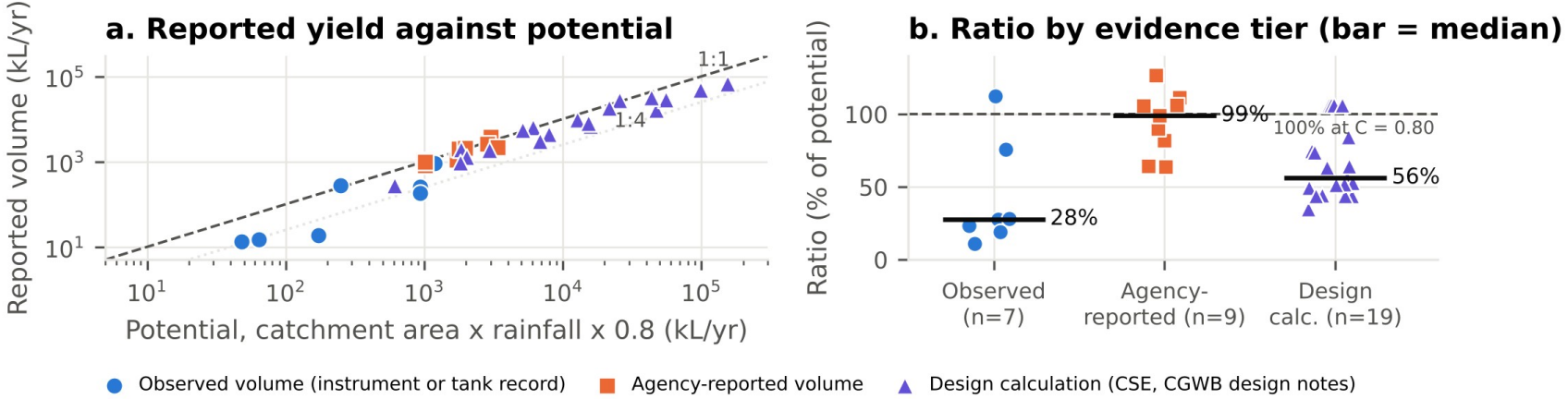
Annual volume stated by CGWB or another agency, no instrument named, or given as a share of estimated runoff. 9 observations, 7 sites.

Design calculation

CSE model projects and one CGWB design note where the 'volume harvested' is a stated share of area x rainfall. 19 sites.

- Screening: 89 site records found (17 not accessible, 7 added from the cost search and the CGWB compilation), 79 screened, 47 excluded (20 potential only, 9 no area or volume, 3 modelled, 10 no roof area, 5 mixed or inconsistent), 35 observations at 31 sites included
- Costs: deflated to 2024 rupees (CPI-IW); 15-year life and O&M 2% of capital are CGWB norms, 8% discount is our assumption (CGWB example: 15%); divided by design yield, then rescaled to observed yield
- Functionality: 13 functional-share estimates from 12 surveys; 90 failure-mode mentions from 40 sources coded into 11 categories

Result 1: in the seven observed cases, yield is about a quarter of potential



Left: reported volume against potential (log scales), 1:1 and 1:4 lines. Right: performance ratio by evidence tier, bars are medians. Observed volumes sit well below agency-reported ones, which cluster at 100% of potential.

Who measured it decides the answer

28%

Observed median, n = 7 at 5 sites. 95% CI 19% to 76%; range 11% to 112%

99%

Agency-reported median, n = 9 at 7 sites. 5 of 9 at or above 99% of potential, the most a roof can give at C = 0.80

56%

Design-calculated median, n = 19. These are the shares CSE assumed, not measurements

Sensitivity checks

Observed median between 25% and 32% for C from 0.90 to 0.70. Dropping the partial-year Goa 2008 observation: 26%. Dropping the two storage-limited houses: 28%.

Observed vs agency-reported: Mann-Whitney U = 10, p = 0.023.

The one borderline case

Warud (112%) counts as observed by the rule (flow meter named) but CGWB gives its volume as a round "90%" of rain received, so it may be an assumed share. Graded agency-reported instead: observed median 26% (n = 6), agency-reported median 102%, p = 0.001.

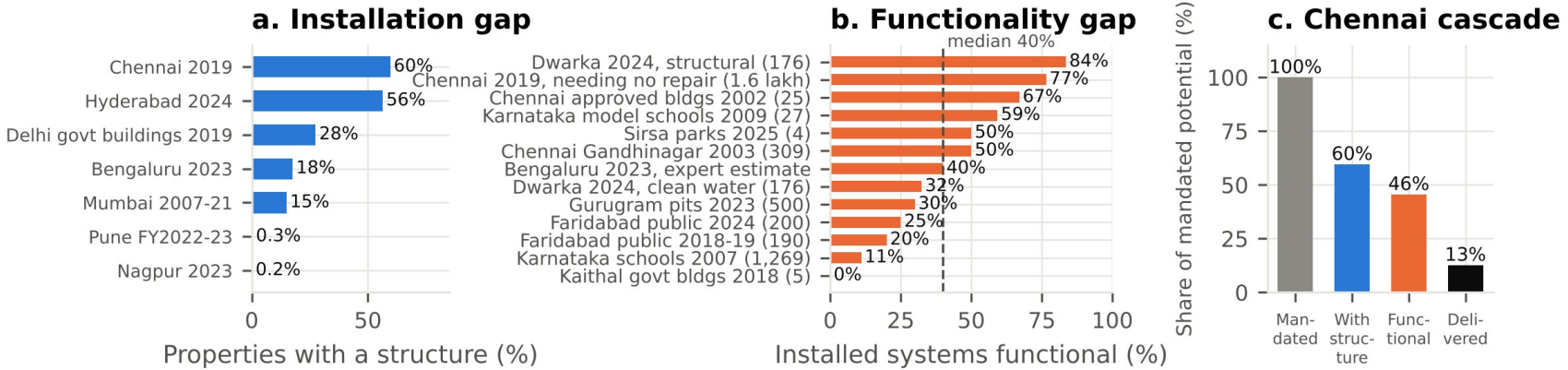
Panjab University's agency figure is stated as "93% of total generated runoff": a design share.

The six midway cases, verified and recomputed

Case	Midway check-in	Final (C = 0.80, stated rainfall)	Status
KITS Ramtek (flow meter)	about 9%	11% of potential; 19 of 215.7 m3 received	Verified in CGWB 2011 p.73; observed
Goa University	61 to 76%	76% (2010); 28% (2008, partial); 19% (2009)	Verified in CSE 2014; observed
BEL Bengaluru	63 to 75%	64% (IMD normal 1,077.6 mm)	Verified; storage, not recharge; agency-reported
Panjab University	about 79%	112% (2,100 m2, 1,059 mm)	Volume is '93% of runoff': design share; agency-reported
Warud (flow meter)	90% (runoff assumption)	112% of potential; 90% of rain received	Verified in CGWB 2011 p.73; observed by rule, agency-reported in sensitivity
IIT Delhi Block VI	not comparable (rainfall NR)	82% with IMD Safdarjung normal	Verified in CGWB 2011 p.44; agency-reported

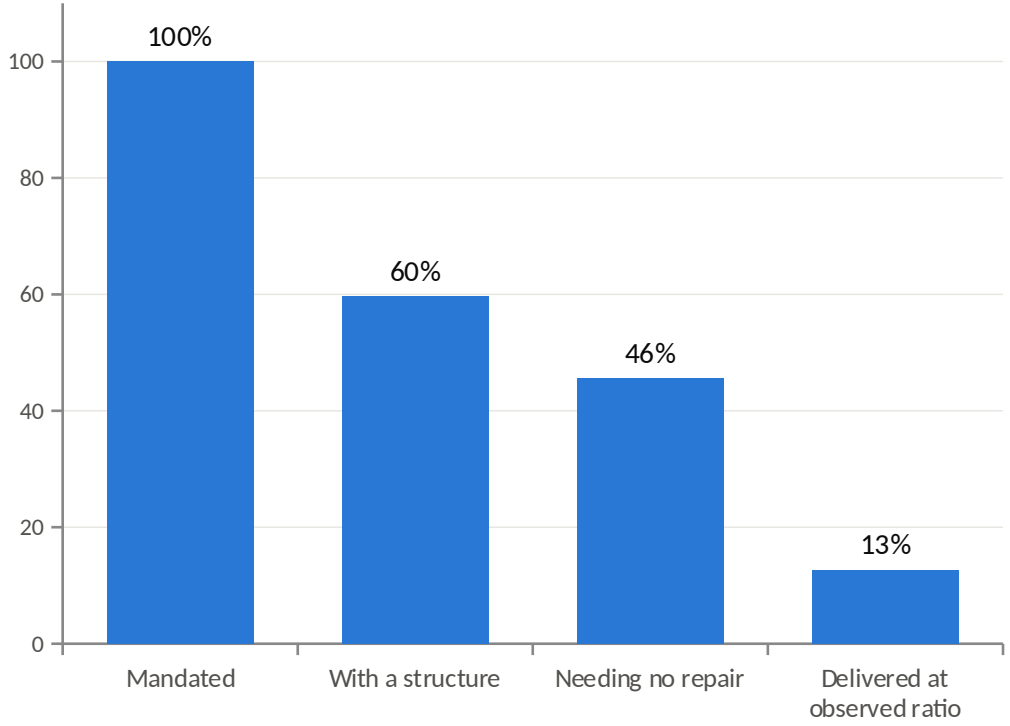
All six cases are kept. With one runoff coefficient and a stated rainfall basis, two cases (Panjab University, Warud) move above 100% of potential; a volume above potential is a design share, not a measurement. Umaid Heritage and PAU Ludhiana stay excluded (mixed catchment, no roof area); Sri Aurobindo Ashram is a design calculation.

Result 2: installed is not the same as working



Installation: 60% of 2.72 lakh houses inspected in Chennai (2019), 56% in Hyderabad, 18% of connected properties in Bengaluru, below 1% in Pune and Nagpur. Functionality: 13 estimates from 12 surveys, median 40% of installed systems working (0 to 84%). Dwarka 2024: 84% structurally intact, only 32% recharging water free of faecal coliform.

Chennai: from mandate to water delivered



60%

of 2.72 lakh inspected houses had a structure (GCC, Aug-Sep 2019 snapshot)

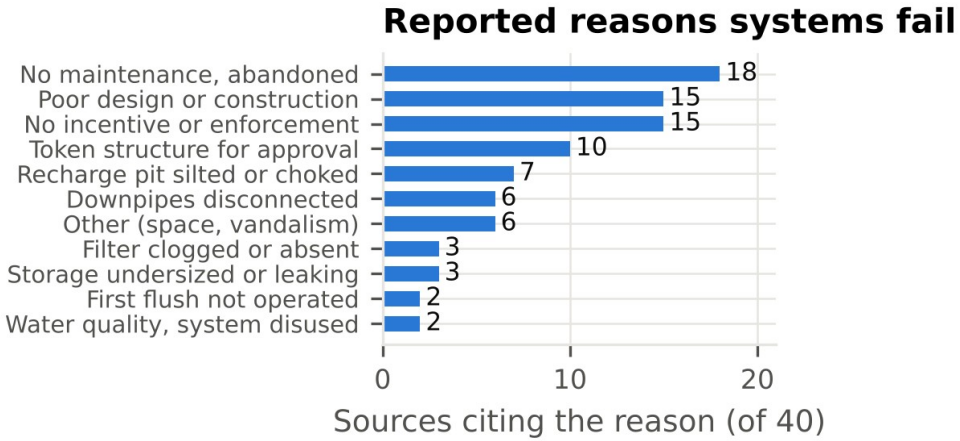
46%

needing no repair: 77% of the 1.62 lakh structures, same inspection

13%

delivered, at the observed yield ratio of 28% (12% with Warud regraded). Bengaluru by the same chain: about 2%

Result 3: why systems fail



90 mentions from 40 surveys, audits and reports, counted by distinct source, 11 categories.

Management causes are cited more often than component faults

No maintenance, abandoned: 18 sources

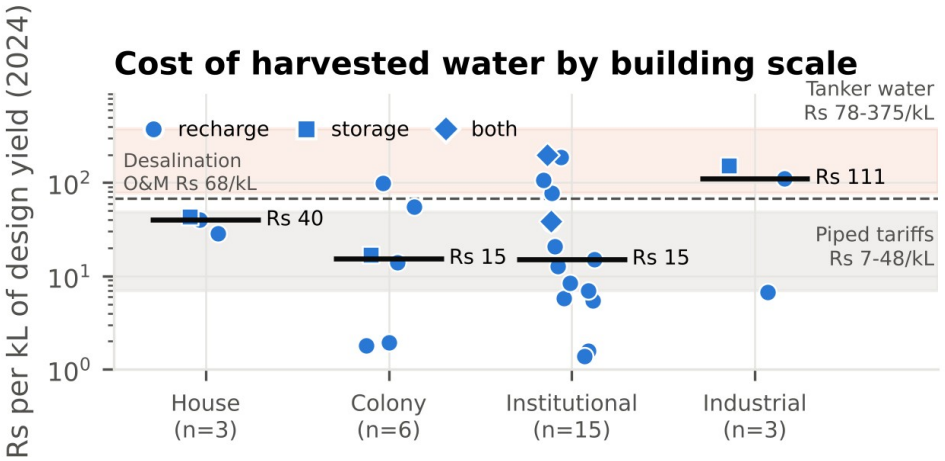
No awareness, incentive or enforcement: 15 sources

Poor design or construction: 15 sources

Token structure, built for compliance only: 10 sources

No maintenance is cited most often; missing incentive or enforcement ties with poor design or construction. Silted pits (7), disconnected downpipes (6) and clogged filters (3) are the faults a routine inspection would catch.

Result 4: cheap at design yield, below tanker prices at observed yield



Rs 21/kL

median levelised cost at design yield, 27 systems (IQR 7 to 66); recharge Rs 15, storage Rs 43

Rs 42/kL

at observed yield (multiplier 2.0): within piped tariffs (Rs 7 to 48), below desalination (Rs 68) and tankers (Rs 78 to 375)

- Cost per kL falls steeply with catchment area (Spearman rho = -0.74, p < 0.001); scale groups not separable at these counts (Kruskal-Wallis p = 0.56)
- Twenty rows are CSE Delhi projects of 2000 to 2004 (median Rs 13); the five post-2010 rows median Rs 98, about Rs 199 at observed yield, inside the tanker range. 2024 rupees, 15-year life, 8% (Rs 29 at 15%), O&M 2%

Unmeasured and under-delivering; reported failures point to enforcement

- The mandates increased installation. Agencies report what they designed; we found only five metered sites, and those deliver about a quarter of potential.
- Structures installed is an output indicator. Water delivered is the outcome. The mandates count structures, not water.
- Reported failures point to maintenance and enforcement first; the technical faults listed are ones an inspection would catch.
- Cost: within piped tariffs at design yield and below tanker prices even at observed yield for the median system. For an owner, recharge still costs more than piped water while the benefit is shared, so incentives matter.

1. Measure

Require a flow meter or recharge-well level logger on systems above a size threshold, so performance can be verified at all.

2. Inspect, then pay

Tie rebates and penalties to a periodic functional inspection, not a one-time completion certificate. Dwarka shows intact structures can recharge contaminated water.

3. Publish

Report installed-versus-functional counts every year, as Chennai did in 2019, so the gap is visible.

Limitations

Small observed tier

7 observations at 5 sites, two of them storage-limited houses and one borderline (Warud). The 28% figure is indicative, not representative. Publication bias favours showcase systems, so the true gap is probably larger. Ramtek's low ratio is partly an intake limit of the dug well.

Heterogeneous inputs

Four observations use an IMD city normal and four a CSE city average, not the year's rainfall. Panjab University's area and volume come from different documents. CSE design calculations use roof plus paved area.

Surveys differ

Definitions range from 'needing no repair' to 'dry pit'; two estimates come from one Dwarka report and one is an expert estimate. Several figures reach us only through press reports; originals cited where found.

Costs and causality

Cost rows are dominated by one programme (CSE Delhi, 2000 to 2004) and depend on the discount rate. Associations are not causal; a weak private incentive to maintain recharge pits is a competing explanation. All of these show why metering is needed.

Contributions and what is submitted

Nilay Shahapure

Analysis (harmonisation, statistics, figures); report, supplementary document, presentation

Vedant Abhangrao

Analysis (screening rules, evidence grading, cost method, reconciliation); video presentation and voice-over

Arjun Jhunjhunwala

Sourcing: locating documents, extracting source records

Yash Krishnani

Sourcing: locating documents, extracting source records

AI tools helped with code, figures, searches and drafting; the statement is in the Supplementary Information.

Submitted

Report (2 pages, IMRaD) and Supplementary Information (15 pages: full tables, screening log, reconciliation with the mid-term report, 168 source URLs)

Data: harmonised dataset for performance, cost, functionality, installation, failure modes and mandates; source extraction tables with a URL and the quoted values for every number

Code: Python scripts that rebuild the dataset, statistics, figures and tables

Every number in the report is traceable to a source URL and the sentence or values taken from the page.

Code repository: github.com/nilay-shahapure/MIRAGE_CM617